

Quiz 4

1 Grading Scheme

- Question 1 is for 1 point. (0.5 for answer and 0.5 for justification)
- Each question 2 is for 1 point. (0.5 for negation and 0.5 for rewriting in English)
- Question 3 is for 2 points. (1 point for computing the size of the sets correctly and 1 point for using the formula correctly)

Total grade is 5

2 Solutions

1. $|\mathcal{P}(A)| = 2^{|A|} = 2^3 = 8$. To construct a subset of A , for each element $x \in A$, we decide if we include or exclude x in the subset. Hence for every $x \in A$, there are two choices and the total number of ways to construct a subset is $2^{|A|}$.
2. (a) $\exists x \in \mathbb{Z}, (x \text{ is even})$. English: There exist an integer x such that x is even
(b) $\exists x \in \mathbb{Z}, (x \geq 0)$. English: There exist an integer x such that x is non-negative.
3. Let $A = \{x \in \mathbb{Z} : 1 \leq x \leq 1000, 2|x\}$, $B = \{x \in \mathbb{Z} : 1 \leq x \leq 1000, 5|x\}$, and $C = \{x \in \mathbb{Z} : 1 \leq x \leq 1000, 10|x\}$. Note that because $\gcd(2, 5) = 10$, we have $A \cap B = C$.

Then, $|A| = 1000/2 = 500$, $|B| = 1000/5 = 200$ and $|C| = 1000/10 = 100$. Number of integers divisible by 2 or 5 $= |A \cup B| = |A| + |B| - |C| = 500 + 200 - 100 = 600$.

3 Common Errors